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PROPOSED GUIDE UACE NAMIREMBE APPLIED MATH MOCK 2026

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Reference File: Namirembe mock S6-math P2.pdf

Subject: Namirembe Mathematics Paper 2

ITEM 1

(a) Determining the Critical Operating Value

Context & Error Correction: The problem states the instability interval is between $x = -1.4$ and $x = -1.2$ for the function $f(x) = 3x^3 - x - 5$. Evaluating the function at these negative values yields

$f(-1.4) = -11.832$

and $f(-1.2) = -8.984$. Since there is no sign change, a root cannot exist in this negative interval. Assuming a standard typographical error in the print, we correct the interval to the positive domain $[1.2,1.4]$ where the physical operating limits exist.

Step 1: Locate the root Evaluate $f(x)$ at the interval boundaries: - $f(1.2) = 3(1.2)^3 - 1.2 - 5 = 3(1.728) - 6.2 = -1.016$ - $f(1.4) = 3(1.4)^3 - 1.4 - 5 = 3(2.744) - 6.4 = 1.832$

Since $f(1.2) < 0$ and $f(1.4) > 0$, a root lies between 1.2 and 1.4.

Step 2: Linear Interpolation for Initial Estimate (x_0) Using the interpolation formula:

$$\frac{x_0 - x_1}{y_0 - y_1} = \frac{x_2 - x_1}{y_2 - y_1}$$

$$x_0 = 1.2 + \frac{0 - (-1.016)}{1.832 - (-1.016)} \times (1.4 - 1.2)$$

$$x_0 = 1.2 + \frac{1.016}{2.848} \times 0.2 \approx 1.2713$$

Step 3: Newton-Raphson Iteration The iterative formula is $x_{n+1} = x_n - \frac{f(x_n)}{f\prime(x_n)}$.
Derivative: $f'(x) = 9x^2 - 1$.

For $x_0 = 1.2713$: - $f(1.2713) \approx -0.101$ - $f'(1.2713) \approx 13.546$ - $x_1 = 1.2713 - \frac{-0.101}{13.546} \approx 1.2788$

For $x_1 = 1.2788$: - $f(1.2788) \approx -0.005$ - $f'(1.2788) \approx 13.718$ - $x_2 = 1.2788 - \frac{-0.005}{13.718} \approx 1.2792$

Since x_1 and x_2 agree to two decimal places, the critical operating value is **1.28**.

(b) Energy Variation in the Filtration Unit

Approximation using the Trapezoidal Rule We are to estimate $\int_1^2 (x-1)\ln x\, dx$ using 6 equal sections. Step size $h = \frac{2-1}{6} = \frac{1}{6}$.

The ordinates (y_n) for $x_n = 1 + n\left(\frac{1}{6}\right)$ are:

- $y_0 = (1-1)\ln 1 = 0$ –

$y_1 = \left(\frac{7}{6} - 1\right)\ln\left(\frac{7}{6}\right) \approx 0.02569$

- $y_2 = \left(\frac{8}{6} - 1\right)\ln\left(\frac{8}{6}\right) \approx 0.09589$ –

$y_3 = \left(\frac{9}{6} - 1\right)\ln\left(\frac{9}{6}\right) \approx 0.20273$

- $y_4 = \left(\frac{10}{6} - 1\right)\ln\left(\frac{10}{6}\right) \approx 0.34055$ –

$y_5 = \left(\frac{11}{6} - 1\right)\ln\left(\frac{11}{6}\right) \approx 0.50511$ –

$y_6 = (2-1)\ln 2 \approx 0.69315$

$$\text{Area} \approx \frac{h}{2}[y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

$$\text{Area} \approx \frac{1}{12}[0 + 0.69315 + 2(1.16997)] \approx 0.25276$$

Analytical Exact Value Using integration by parts: $\int u\, dv = uv - \int v\, du$. Let $u = \ln x \Longrightarrow du = \frac{1}{x}dx$. Let $dv = (x-1)dx \Longrightarrow v = \frac{x^2}{2} - x$.

$$\int_1^2 (x-1)\ln x\, dx = \left[\left(\frac{x^2}{2} - x\right)\ln x\right]_1^2 - \int_1^2 \left(\frac{x}{2} - 1\right) dx$$

$$= \big((2-2)\ln 2 - (-0.5)\ln 1\big) - \left[\frac{x^2}{4} - x\right]_1^2$$

$$= 0 - \left(\left(\frac{4}{4} - 2\right) - \left(\frac{1}{4} - 1\right)\right) = -\big(-1 - (-0.75)\big) = 0.25$$

Discrepancy and Reliability Discrepancy $= |0.25276 - 0.25| = 0.00276$. ***Since*** $\mathbf{0.00276 \leq 0.003}$***, the students' approximation method is reliable. How to improve reliability: The approximation can be improved by increasing the number of strips/sections (i.e., reducing the step size*** h***)***

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ITEM 2

Data Organization We convert the cumulative frequency into a grouped frequency distribution.

Mass (Kg)	Freq (f)	Midpoint (x)	fx	fx^2	Width	Freq Density
45 - 50	9	47.5	427.5	20306.3	5	1.8
50 - 60	16	55.0	880.0	48400.0	10	1.6
60 - 75	21	67.5	1417.5	95681.3	15	1.4
75 - 80	17	77.5	1317.5	102106.3	5	3.4
80 - 85	14	82.5	1155.0	95287.5	5	2.8
85 - 90	5	87.5	437.5	38281.3	5	1.0
90 - 100	7	95.0	665.0	63175.0	10	0.7
100 - 105	4	102.5	410.0	42025.0	5	0.8
105 - 110	1	107.5	107.5	11556.3	5	0.2
Total	**94**		**6817.5**	**516818.8**		

1. Most Common Body Mass (Mode) The modal class is $75 - 80$ (highest frequency density of 3.4). Using the mode formula:

$$\text{Mode} = L_1 + \left(\frac{D_1}{D_1 + D_2}\right)c$$

Where: $L_1 = 75$,

$D_1 = 3.4 - 1.4$

$= 2.0,$

$D_2 = 3.4 - 2.8$

$= 0.6,$

$c = 5.$

$$\text{Mode} = 75 + \left(\frac{2.0}{2.6}\right) \times 5$$

$$= 75 + 3.846$$

$$= 78.85 \text{ kg}$$

2. Average Body Mass (Mean)

$$\bar{x} = \frac{\sum fx}{\sum f}$$

$$= \frac{6817.5}{94} \approx 72.53 \text{ kg}$$

3. Measure of Variation (Standard Deviation)

$$\text{Variance} = \frac{\sum fx^2}{\sum f} - (\bar{x})^2$$

$$\text{Variance} = \frac{516818.75}{94} - (72.5266)^2$$

$$= 5498.07 - 5260.11$$

$$= 237.96$$

$$\text{Standard Deviation} = \sqrt{237.96} \approx 15.43 \text{ kg}$$

Conclusion on Nutritional Outcomes The feeding programme produces balanced outcomes if: - Mode and Mean are between 60 kg and 70 kg. - Standard Deviation ≤ 10 kg.

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Findings: - The Mode (78.85 kg) and Mean (72.53 kg) both exceed the 70 kg upper threshold. - The Standard Deviation (15.43 kg) exceeds the 10 kg limit, indicating too wide a spread. Report Verdict: The feeding programme is NOT producing balanced nutritional outcomes. There is a clear tendency towards overweight masses and highly inconsistent bodily growth among the finalists.

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ITEM 3

Part 1: Road Accident Investigation

Let A be the event of having an accident, and G be the event of wearing glasses. - Total drivers: $n(S) = 30$

- Drivers in accidents: $n(A) = 9$

- Non-accident drivers: $n(A') = 30 - 9 = 21$

- Accident victims wearing glasses: $n(A \cap G) = 5$

Given that the probability of wearing glasses among those with no accidents is $\frac{1}{3}$:

$$P(G|A') = \frac{1}{3} \Longrightarrow n(A' \cap G) = \frac{1}{3} \times 21 = 7$$

Total number of drivers wearing glasses:

$$n(G) = n(A \cap G) + n(A' \cap G) = 5 + 7 = 12$$

Objectives Check: 1. **Likelihood randomly selected driver wears glasses:**

$$P(G) = \frac{12}{30} = 0.4 \text{ (or 40\%)}$$

2. **Is wearing glasses common among accident victims?** The probability that an accident victim wears glasses is:

$$P(G|A) = \frac{n(A \cap G)}{n(A)} = \frac{5}{9} \approx 0.5556 \text{ (or 55.56\%)}$$

Since $55.56\% > 45\%$, the researchers conclude that wearing glasses **does appear common** among accident victims.

Part 2: University Wellness Records

Let height $X \sim N(\mu, \sigma^2)$. We are given: - $P(X > 200) = 0.01 \Longrightarrow P(X \leq 200) = 0.99$ - $P(165 < X < 200) = 0.76$

From standard normal tables, the Z-value for an area of 0.99 is $z_1 \approx 2.326$.

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